# Jacobsthal Coverings: Exact Stopped-Tree Certificates, Joint Moments, and Quantitative Regeneration Errors

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**Abstract**

We analyze the finite covering and probabilistic comparison mechanisms in *A quadratic bound for Jacobsthal's function*. Exact residue-class counts, parity-sensitive tree bounds, common-law moments and regeneration errors are treated separately from the global arithmetic estimates. The report develops finite certificates and local refinements with explicit hypotheses and executable checks. A finite triangular identity bounds the variance of a consecutive random-sieve count by its mean, with stable measure errors and a nonunit-progression correction. In addition, exponential cycle information yields an explicit marked-window error bound without assuming independence of marks and their cycle lengths. Imported sieve, inverse, prime-transfer and stopping estimates remain identified as such; finite verification does not prove a uniform quadratic bound. Contribution assessment: C3/5, finite variance and stability bounds and quantitative marked-window errors significantly sharpen local source inputs while the global arithmetic transfer remains imported.

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# 1 The Arithmetic Claim and the Local Task

For a positive integer $n$, $j(n)$ is the least positive length such that every interval of that many consecutive integers contains an integer coprime to $n$. Put

$$h(k) = \sup_{\omega(n)\le k} j(n), \qquad k \ge 1,$$

where $\omega(n)$ counts distinct prime divisors. The source claims $h(k) \ll k^2/(\log\log(3k))^2$, uniformly in the sizes of the primes and in the interval's translation. This is not merely a primorial assertion, and the implicit constant is not made explicit. We do not independently certify that global theorem.

For distinct primes $p$ and prescribed residues $a_p$, the Chinese remainder theorem gives $b \equiv a_p \pmod p$ simultaneously. Avoiding those classes at an integer $m$ is exactly $\gcd(m-b, \prod p) = 1$. Thus a finite certificate for one assignment proves a statement about its corresponding translates, not automatically about every assignment or every collection of primes.

The source uses, for sufficiently large $z$,

$$L = \log z, \quad Y = \lfloor z^2/L^2 \rfloor, \quad w = L/(\log L)^2, \quad B = L/\log w, \quad V_0 = \prod_{p\le w}(1-1/p).$$

Its main intermediate assertion is that every prescribed class system at all primes $p \le z$ leaves at least $cYV_0/B^2$ survivors in $[1, Y]$. To pass to at most $k$ arbitrary divisor primes it takes $z = A_0 k \log k/\log\log k$ and bounds the survivors removed by each remaining prime using a progression upper sieve. This additional step cannot be replaced by checking the first $k$ primes.

The leading lower linear-sieve function vanishes at parameter 2. The source deliberately retains a boundary correction at scale $B^{-2}$. Its starting continuation has the negative contribution $-a_\star e^\gamma B^{-2} + o(B^{-2})$, with $a_\star = 1/100$. Discarding this sign, or proving positivity only for a reference tree, would not prove the claimed survivor estimate.

# 2 Exact Finite Tree Coordinates

Fix an integer $Y \ge 1$, a finite small-prime set $\mathcal{P}_0$, and a finite disjoint ordered set $\mathcal{P}_1$ of larger primes. Fix one forbidden residue at each of these primes. No asymptotic size assumptions are needed in this section. A node is a strictly decreasing tuple of primes from $\mathcal{P}_1$, with squarefree product $d$ and depth $|d|$. Its available primes are the smaller unused primes, with all $\mathcal{P}_1$ available at the root. Let $N_d$ count integers in $[1, Y]$ hitting the classes at every factor of $d$ and avoiding $\mathcal{P}_0$. Let $S_d$ additionally avoid every available prime.

**Lemma 2.1** (Exact child partition)**.** *For every node,*

$$S_d = N_d - \sum_{p \text{ available}} S_{dp}.$$

*Proof.* An integer counted by $N_d$ but not $S_d$ has a unique smallest available prime at which it hits a forbidden class. It belongs to the corresponding child $dp$, whose survivor condition avoids all smaller available primes. The child sets are disjoint and exhaust the difference. The fact that the partition uses the smallest bad prime is consistent with decreasing prime tuples. □

The source expands every child at an even node, but may omit children at an odd node. Its particular admission rule is $r - x \geq \max(2x, x+2)$, where $r = \log_w(Y/d) - a_\star + 2$ and $x = \log_w p$. Its purpose is to keep included progressions long enough for a uniform absolute upper sieve. The sign argument itself permits any declared odd-node subset.

**Proposition 2.2** (Stopped comparison, finite form)**.** *Assign reference numbers $\mu_d$ to the nodes of a finite included tree. Define its reference subtree recursively using the same included children. Let $\mathcal{S}$ be a prefix-free set of even stops and let $\mathcal{T}$ be the retained expanded nodes, excluding stops. Then its stopped lower expression is*

$$L_{\mathrm{stop}} = \sum_{d\in\mathcal{T}} (-1)^{|d|} N_d + \sum_{d\in\mathcal{S}} S_d \leq S_1.$$

*If $R_d$ is the full reference subtree value at $d$, then*

$$L_{\mathrm{stop}} - R_1 = \sum_{d\in\mathcal{T}} (-1)^{|d|}(N_d - \mu_d) + \sum_{d\in\mathcal{S}} (S_d - R_d). \tag{1}$$

*Proof.* At an even node subtracting upper child bounds gives a lower bound. At an odd node, omitted child counts are nonnegative, so omitting them and subtracting lower bounds gives an upper bound. Induct upward in the finite tree. Exact counts at even stops are legitimate lower bounds there. Prefix-freeness prevents duplicate expansion. Expanding the full reference expression to exactly the same stops and subtracting proves the identity. □

For the source, $\mu_d = (Y/d)V_0$ and $R_d = \mu_d P_i(r_d, b_d)$ in the declared parity. The total prefix harmonic mass has the elementary majorant $\prod_{p\in\mathcal{P}_1}(1+1/p)$; this finite identity comes from expanding the product over subsets. Its asymptotic $O(B)$ bound uses reciprocal-prime estimates. Neither is a probability normalization.

## 3 Finite Random-Sieve Variance Without a Limit

Here the probability space is precisely the product of uniform forbidden classes, not a distribution on the actual isolated primes. Let $\mathcal{P}$ be any finite set of primes, $W = \prod_{p\in\mathcal{P}} p$ and $V = \prod_{p\in\mathcal{P}}(1-1/p)$. Choose $a_p$ independently and uniformly modulo $p$, put $I_j = \mathbf{1}_{\{j\not\equiv a_p \pmod p\ \forall p\in\mathcal{P}\}}$, and let $S = \sum_{j=u}^{u+J-1} I_j$, where $u \in \mathbb{Z}$ and $J \geq 1$ is an integer. Set $\kappa = 2$ if $2 \in \mathcal{P}$ and $\kappa = 1$ otherwise, and $W_{\mathrm{odd}} = \prod_{p\in\mathcal{P},\ p>2} p$. For squarefree $\ell \mid W_{\mathrm{odd}}$, define

$$\lambda_\ell = \prod_{p\mid\ell} \frac{1}{p-2}, \qquad C = \prod_{\substack{p\in\mathcal{P}\\ p>2}} \left(1 - \frac{1}{(p-1)^2}\right).$$

Empty products equal one.

**Theorem 3.1** (Exact triangular deficit)**.** *Write $J = \kappa\ell n_\ell + r_\ell$, where $n_\ell = \lfloor J/(\kappa\ell)\rfloor$ and $0 \leq r_\ell < \kappa\ell$. Then $\mathbb{E}S = JV$ and*

$$\operatorname{Var} S = JV - (JV)^2 C \sum_{\ell\mid W_{\mathrm{odd}}} \lambda_\ell \frac{r_\ell^2 + \kappa^2\ell^2 n_\ell}{\ell J^2} \leq JV. \tag{2}$$

*Consequently, for every $\eta > 0$,*

$$\mathbb{P}(|S - JV| > \eta JV) \le \frac{1}{\eta^2 JV}.$$

*Proof.* One forbidden class excludes a fixed position with probability $1/p$. For distinct positions at spacing $h$, it excludes one residue if $p \mid h$ and two otherwise. Thus, exactly as in source `variance.tex`, equation `variance-pair-factor`,

$$\frac{\mathbb{E} I_i I_{i+h}}{V^2} = \kappa \mathbf{1}_{\kappa \mid h}\, C \sum_{\substack{\ell \mid W_{\mathrm{odd}} \\ \ell \mid h}} \lambda_\ell.$$

If $2 \in \mathcal{P}$, the parity zero is exact; otherwise the parity factor is one. Expanding the finite product is legitimate without a convergence theorem. The normalized off-diagonal second moment is

$$\frac{\mathbb{E}[S(S-1)]}{(JV)^2} = C \sum_{\ell \mid W_{\mathrm{odd}}} \lambda_\ell A_\ell, \qquad A_\ell = \frac{2\kappa}{J^2} \sum_{\substack{1 \le h < J \\ \kappa\ell \mid h}} (J - h).$$

Including $h = J$ contributes zero. Therefore

$$A_\ell = \frac{2\kappa n_\ell J - \kappa^2 \ell n_\ell (n_\ell + 1)}{J^2}, \qquad \frac{1}{\ell} - A_\ell = \frac{r_\ell^2 + \kappa^2 \ell^2 n_\ell}{\ell J^2} \ge 0.$$

This covers $n_\ell = 0$ as well. Finally,

$$C \sum_{\ell \mid W_{\mathrm{odd}}} \frac{\lambda_\ell}{\ell} = \prod_{\substack{p \in \mathcal{P} \\ p > 2}} \left(1 - \frac{1}{(p-1)^2}\right)\left(1 + \frac{1}{p(p-2)}\right) = 1.$$

Subtract $(JV)^2$ from $\mathbb{E} S^2 = JV + \mathbb{E}[S(S-1)]$ to obtain (2). Markov's inequality applied to the squared deviation proves the tail bound. □

The parent review supplied the finite-triangular refinement and its odd-prime extension; the argument above independently derives it from the source pair formula. It is a quantitative extraction from classical two-point singular-series algebra, not a claim of historical novelty. For the empty prime set, $\kappa = C = V = W_{\mathrm{odd}} = 1$ and the formula gives variance zero. The source's dominated-convergence proof is valid, but unnecessary for this upper bound. For primes at most $w$, the sole variance requirement for relative concentration is $JV \to \infty$; by Mertens' theorem this is equivalent to $J/\log w \to \infty$. In particular the source range $J \ge w^\delta$, fixed $\delta > 0$, is covered. No uniform arithmetic transfer to this enlarged range is asserted.

**Proposition 3.2** (Stable positive-measure ledger)**.** *Let $\mu$ be a finite positive measure with indicators $I_j$ on a consecutive interval of $J$ indices. Write $P_{ij}$ for the pair probabilities in Theorem 3.1. Suppose $\epsilon_i \ge 0$ and*

$$|\mu(\Omega) - 1| \le \epsilon_0, \quad \left|\int I_j \, d\mu - V\right| \le \epsilon_1 V, \quad \int I_i I_j \, d\mu \le (1 + \epsilon_2) P_{ij} \quad (i \ne j).$$

*In particular $P_{ij} = 0$ requires a zero integral. Then*

$$\int (S - JV)^2 \, d\mu \le (1 + \epsilon_1) JV + (\epsilon_2 + 2\epsilon_1 + \epsilon_0)(JV)^2. \tag{3}$$

*If an actual probability law $\nu$ satisfies $\nu(E) \le K\mu(E) + \delta$ for every event, with $K, \delta \ge 0$, then*

$$\nu(|S - JV| > \eta JV) \le \frac{K}{\eta^2}\left\{\frac{1 + \epsilon_1}{JV} + \epsilon_2 + 2\epsilon_1 + \epsilon_0\right\} + \delta.$$

*Proof.* The diagonal sum is at most $(1+\epsilon_1)JV$ and the off-diagonal sum is at most $(1+\epsilon_2)(JV)^2$ by the preceding theorem. Use the lower bound $\int S \geq (1-\epsilon_1)JV$ and upper bound $\mu(\Omega) \leq 1+\epsilon_0$ in the exact expansion $\int S^2 - 2JV \int S + (JV)^2\mu(\Omega)$. Positivity then permits Markov's inequality without normalizing $\mu$. The assumed domination gives the last assertion. □

### 3.1 Progressions and the Nonunit Obstruction

Theorem 3.1 holds unchanged on $u + a\{0, \ldots, J-1\}$ whenever $\gcd(a, W) = 1$: the transformed classes $(a_p - u)a^{-1} \pmod p$ are still independent and uniform. It is false for arbitrary probes. On $\{0, W, \ldots, (J-1)W\}$ all indicators coincide, so $\operatorname{Var} S = J^2V(1-V)$.

More generally let $\mathcal{D} = \{p \in \mathcal{P} : p \mid a\}$ and $v_{\mathcal{D}} = \prod_{p\in\mathcal{D}}(1-1/p)$. The survival event $B$ at those primes has probability $v_{\mathcal{D}}$. It is independent of the remaining transformed model, whose count is $S'$, and $S = \mathbf{1}_B S'$. Hence the exact decomposition

$$\operatorname{Var} S = v_{\mathcal{D}} \operatorname{Var} S' + v_{\mathcal{D}}(1-v_{\mathcal{D}})(\mathbb{E}S')^2 \tag{4}$$

contains an unavoidable between-component term. Apply Theorem 3.1 to the remaining primes, whether or not they contain 2. Since $\mathbb{E}S' = JV'$ and $V = v_{\mathcal{D}}V'$, it gives

$$\operatorname{Var} S \leq JV + \frac{1-v_{\mathcal{D}}}{v_{\mathcal{D}}}(JV)^2.$$

When all primes divide the step, $S' = J$ is deterministic and (4) recovers the exact counterexample variance.

### 3.2 What Still Requires Arithmetic

In source `variance.tex`, $H = Dq_{f,E}$, $C_0 = (Db - A)/q_{f,A}$ and $T = \max(H, |C_0|/R)$. Reducedness and nonalignment give $(C_0, H) = 1$. The hypothetical aligned progression is parametrized by

$$m_0 \in [C_0/p, C_0/p + H), \quad pm_0 \equiv C_0 \pmod H, \qquad Dn - A = pq_{f,A}(m_0 + Hj).$$

Its uniqueness matters when counting pairs instead of primes. For $H \leq R^{4/5}$, the source combines Brun–Titchmarsh with a dimension-two fundamental lemma to build an unnormalized dominating measure. For larger $H$, modular-hyperbola completion and an upper sieve dominate full residue patterns. These are not consequences of Theorem 3.1; the actual prime classes are never assumed iid. Our finite ledger may replace the random-model estimate once these domination and moment hypotheses have been established.

The source freezes a common integer interval with slack $J \geq w^{a_\star/2+\delta_0}$. Its $O(\xi J + 1)$ boundary displacement costs $O(\xi JV_0)$ only after the progression upper sieve is applied, with fixed $\xi > 0$ before $z \to \infty$. Weakening the random-model length assumption does not remove that separate boundary obligation.

## 4 Joint Moments and Signed Certificate Stability

The private originals consulted for this section are the common-law covariance and remainder proofs in `TR-core-pat-16`, lines 184–328, and the local-moment chapter of `volume1-rev2`, lines 14–226. They are mathematical inputs for the formulations below, not evidence of an arithmetic sieve theorem. Their contents were read locally; neither private text nor private titles were submitted to public searches.

Let $Q$ be a probability measure on complete residue-class configurations. For finitely many node counts or edge defects, form a real vector $Z = (Z_1, \dots, Z_m)$ in $L^2(Q)$. Define

$$m = \mathbb{E}_Q Z, \quad T = \mathbb{E}_Q Z Z^{\mathsf{T}}, \quad M = T - mm^{\mathsf{T}}.$$

These are raw and centered moments under the same law. For every $c \in \mathbb{R}^m$, $c^{\mathsf{T}} M c = \mathbb{E}_Q[c^{\mathsf{T}}(Z - m)]^2 \geq 0$. The source's retained signed error is a linear functional of such defects. Its variance is $c^{\mathsf{T}} M c$, not merely $\sum c_i^2 M_{ii}$. The latter discards cross-node information. When one uses polynomial features of degree $d$, second moments of the feature vector require moments through degree $2d$.

**Proposition 4.1** (A signed remainder certificate)**.** *Suppose $Z = G + R$ on the same probability space and $\mathbb{E}\,\|R\|^2 \leq \epsilon^2$. Put $V_G = \mathbb{E}\,\|G - \mathbb{E}G\|^2$. Then*

$$\|\mathbb{E}Z - \mathbb{E}G\| \leq \epsilon, \qquad \|\operatorname{Cov} Z - \operatorname{Cov} G\|_{\mathrm{op}} \leq 2\sqrt{V_G}\epsilon + \epsilon^2.$$

*More particularly, for each fixed $c$,*

$$\sqrt{\operatorname{Var}(c^{\mathsf{T}} Z)} \leq \sqrt{c^{\mathsf{T}} \operatorname{Cov}(G) c} + \|c\|\,\epsilon. \tag{5}$$

*Proof.* Jensen's inequality gives the mean bound. Centering is an orthogonal projection in $L^2$, so $\mathbb{E}\,\|R - \mathbb{E}R\|^2 \leq \epsilon^2$. Expand the centered outer product of $G + R$. Each cross term has operator norm at most $\sqrt{V_G}\epsilon$ by Cauchy–Schwarz; the remainder covariance has operator norm at most $\epsilon^2$. For the directional statement apply the $L^2$ triangle inequality to the two centered scalar variables. □

This standard common-law result is reproduced to specify a usable certificate, not presented as a newly discovered covariance theorem. If only componentwise errors $\mathbb{E}R_i^2 \leq \epsilon_i^2$ are known, the final term in (5) may instead be $\sum_i |c_i|\epsilon_i$, often smaller than $\|c\|\,(\sum_i \epsilon_i^2)^{1/2}$. Both follow without independent approximation errors. For example, if $Z_1 = Z_2$, the difference $Z_1 - Z_2$ has zero variance, whereas an independent-copy substitution usually has positive variance.

There is also a useful interval decomposition. If $U = \sum_{\alpha=1}^m c_\alpha S_\alpha$, where $S_\alpha$ counts survivors on a consecutive interval of length $J_\alpha$ under the same model of Theorem 3.1, then

$$\operatorname{Var} U \leq V \left( \sum_{\alpha=1}^m |c_\alpha| \sqrt{J_\alpha} \right)^2. \tag{6}$$

This follows by applying the $L^2$ triangle inequality to centered interval counts. Intervals may overlap. This extends the useful bound to step weights without falsely asserting $\operatorname{Var} U \leq V \sum c_\alpha^2 J_\alpha$.

### 4.1 Exact Finite Covering Certificates

For fixed primes and a fixed class assignment, a covering certificate lists, for every $m \in [1, Y]$, one prime $p$ with $m \equiv a_p \pmod p$. Checking it proves $j(\prod p) > Y$. A survivor list or the exact stopped lower expression proves noncoverage for that assignment. To bound $j(\prod p)$ above, every residue assignment must be covered by the verification, or all translates modulo $\prod p$ must be checked. For this fixed prime set, CRT makes these two exhaustive tests equivalent.

A portable stopped certificate contains the included child sets, prefix-free even stops, the integers $N_d, S_d$, reference rationals $\mu_d, R_d$ and the two sides of (1). All entries can be checked with integer congruences and rational arithmetic. Omitting an even child is not an admissible shortcut. Prefix-freeness and the declared odd-child subsets are part of the certificate, not informal implementation details. The executable checks supplied with this report test these identities against direct survivor enumeration and exercise different odd-node admission choices and stop choices. Their purpose is to detect sign, normalization and bookkeeping errors at finite size, not to certify the infinite family of arithmetic comparisons.

## 5 The Tilted Kernel and Compact Approximation

The source sets $A = 2e^{\gamma}$, with linear-sieve functions $f, F$ satisfying $(sf(s))' = F(s-1)$ and $(sF(s))' = f(s-1)$ on their respective delay ranges. Write $\phi_{\mathrm{e}}(s) = s^2 f'(s)$ and $\phi_{\mathrm{o}}(s) = -s^2 F'(s)$. The source proves positivity and the future-integral identities

$$\phi_i(s) = \int_{m_i(s)}^{\infty} \frac{t+1}{t^2} \phi_{i'}(t)\, dt, \qquad K_i(s, dt) = \frac{(t+1)\phi_{i'}(t)}{t^2 \phi_i(s)} \mathbf{1}_{t \geq m_i(s)}\, dt, \tag{7}$$

where $m_{\mathrm{e}}(s) = s-1$ and $m_{\mathrm{o}}(s) = \max(2, s-1)$. Normal even states have $s \geq 2$ and odd states $s \geq 1$. The source also treats a bounded initial even extension below 2; it is not an extension of an omitted child's lower-sieve value. Formula (7) is a row normalization at each starting state, not a statement that Lebesgue measure is invariant.

If the current gap is $r$, then the next exponent is $x = r/(t+1)$ and the next gap is $r' = rt/(t+1)$. The one-step harmonic element is $dx/x = dt/(t+1)$ after reversing orientation. Consequently

$$\frac{dt/(t+1)}{K_i(s, dt)} = \left(\frac{r'}{r}\right)^2 \frac{\phi_i(s)}{\phi_{i'}(t)}. \tag{8}$$

Along a fixed-length path the factors telescope to $(r_n/r_0)^2 \phi_{i_0}(s_0)/\phi_{i_n}(s_n)$. This exact change of measure explains both the root scale $B^{-2}$ and the inverse-$\phi$ occupation rewards. A numerical probability kernel that omits the endpoint factor does not approximate the harmonic path measure.

The raw invariant densities identified in the source are $\pi_{\mathrm{e}}(s) = \phi_{\mathrm{e}}(s)$, $s \geq 2$, and $\pi_{\mathrm{o}}(s) = (1 - s^{-2})\phi_{\mathrm{o}}(s)$, $s \geq 1$. They are not normalized probability densities. Its occupation normalization is

$$M_g = \sum_i \int \pi_i(s) g(s)\, ds, \qquad g(s) = \log(1 + 1/s),$$

rather than $\sum_i \int \pi_i$. The parity alternation is retained.

**Proposition 5.1** (Positive compact kernel compression)**.** *Fix a state $(i, s)$ in (7), a finite interval $D = [m_i(s), T]$ and a partition into intervals of width at most $h$. Let $\tau = K_i(s, D^c)$. In each cell of positive mass $q_j$ use the conditional mean $t_j = q_j^{-1} \int_{cell} t\, K_i(s, dt)$, and put $\widehat{K} = \sum_j q_j \delta_{t_j}$. If $H$ is twice continuously differentiable on $D$, then*

$$\left| \int_D H\, dK_i - \int H\, d\widehat{K} \right| \leq \frac{h^2}{8}(1 - \tau) \sup_D |H''|. \tag{9}$$

*For bounded $H$ on the full support, restoring an arbitrary tail atom of mass $\tau$ costs at most an additional $2\tau \|H\|_\infty$. The compact conditional probability $\widehat{K}/(1-\tau)$, when $\tau < 1$, is different from the killed kernel $\widehat{K}$.*

*Proof.* The first-order Taylor term at $t_j$ integrates to zero. The remainder is bounded by $\frac{1}{2} \sup |H''|(t - t_j)^2$. For any probability law on $[a, b]$, $\mathbb{E}(X - \mathbb{E}X)^2 \leq (b-a)^2/4$: use $(X-a)(b-X) \geq 0$ and maximize $(\mathbb{E}X - a)(b - \mathbb{E}X)$. Sum this estimate with weights $q_j$. The true tail integral and an arbitrary tail atom each have absolute value at most $\tau \|H\|_\infty$. Their difference gives the stated cost. The mass of the killed kernel is $1 - \tau$, proving the last distinction. □

The conditional-mean construction preserves the constant and first moment exactly on every cell, uses positive weights, and gives a second-order error for the actual next-state observables. For example $g''(t) = (2t+1)/(t^2(t+1)^2)$, so its supremum on any positive compact interval is finite and

computable. For a reward involving $1/\phi_{i'}(t)$ the derivative bound is needed only on that chosen compact interval. No globally uniform bound is inferred as $\phi_{i'}(t)$ tends to zero. Using midpoints instead of conditional means generally loses the first-order cancellation. Evaluating unknown cell moments by uncertified floating point quadrature does not itself produce a rigorous certificate.

For discontinuous admission tests there is a separate boundary ledger. If a cellwise representative is within $h$ of the original state and a test is $\mathbf{1}_{t\le c}$, only the cell meeting $c$ can change the test; its error is bounded by $K_i(s,[c-h,c+h])$. More generally, a Lipschitz test function $b$ with constant $L_b$ can change the sign of $b$ only where $|b|\le L_b h$. This is a boundary-mass estimate, not a Taylor estimate for an indicator. Products of compressed transitions also need pathwise accumulation of these costs. We claim a one-step local improvement, not automatic renewal convergence or uniform prime coupling for the compressed chain.

## 6 Exponential Marked-Window Errors

The continuous chain in the source regenerates at odd states $s\le 3$, before taking the next draw. At those states the outgoing law is the same, since $m_{\mathrm{o}}(s)=2$ and $\phi_{\mathrm{o}}(s)=A$. The complete regeneration cycles are iid objects. If $G$ is their total logarithmic gap decrement, the source proves

$$G\ge g_*=\log(4/3)>0,\qquad \mathbb{E}e^{\theta_0 G}<\infty$$

for some $\theta_0>0$. It also proves a uniform exponential moment for the time to the first regeneration from the allowed initial compact ratio range. The increments $g(s)$ are bounded in those initial and normal ranges, so the initial logarithmic decrement $D_0$ has a uniform exponential moment for a possibly smaller parameter. These are inputs established in the source's regeneration argument. The result below is proved abstractly from precisely these inputs.

Mark a cycle if its first draw is in $[2.09,2.13]$. The mark has probability $p\in(0,1)$ independent of the initial gap. It need not be independent of $G$ within the same cycle. The source uses finite mean marked spacings to obtain a qualitative window estimate. The next theorem extracts an exponential estimate from the stronger cycle hypotheses already available.

**Theorem 6.1** (A quantitative delayed marked renewal bound)**.** *Let $(G_j,B_j)_{j\ge0}$ be iid pairs, with $G_j\ge g_*>0$, $B_j\in\{0,1\}$ and $\mathbb{P}(B_j=1)=p\in(0,1)$. Suppose $D_0\ge0$ is independent of these pairs and that for some $\theta_0>0$, $\mathbb{E}e^{\theta_0 G_0}<\infty$ and $\mathbb{E}e^{\theta_0 D_0}\le M_0<\infty$. For sufficiently small $\theta>0$ put*

$$m_b=\mathbb{E}[e^{\theta G_0}\mathbf{1}_{B_0=b}],\quad m_0<1,\quad M_W=\frac{m_1}{1-m_0},\quad M_D=\frac{p\,\mathbb{E}e^{\theta D_0}}{1-m_0}.$$

*Let $\mathcal{R}$ be the set of beginnings of marked cycles, with time measured from the original start including $D_0$. For all $v,H\ge0$,*

$$\mathbb{P}(\mathcal{R}\cap[v,v+H]=\varnothing)\le M_D e^{-\theta v}+\frac{C_* M_W}{1-e^{-\theta}}e^{-\theta H},\qquad C_*=2+\lfloor 1/g_*\rfloor. \tag{10}$$

*No nonarithmetic hypothesis is required for this upper bound.*

*Proof.* Continuity at $\theta=0$ gives $m_0\to 1-p<1$, justifying the choice. Starting at a marked beginning, the next marked beginning occurs after one marked cycle followed by a geometric number of unmarked cycles. Summing their exponential transforms gives

$$\mathbb{E}e^{\theta W}=\frac{m_1}{p}\sum_{n\ge0}p\,m_0^n=M_W.$$

The first marked beginning occurs after $D_0$ followed by unmarked cycles, with transform $\mathbb{E}e^{\theta D} = \mathbb{E}e^{\theta D_0}\sum_{n\ge 0} pm_0^n = M_D$. The iid-pair assumption implies that successive marked spacings are iid and independent of the delay: the cycle at a marked beginning is reserved for the subsequent spacing. This convention prevents using a marked cycle twice or wrongly replacing its conditional length distribution by the unconditional one.

For ordinary marked renewals starting at zero, let $U = \sum_{n\ge 0} \mathcal{L}(W_1 + \cdots + W_n)$, including the atom at zero. Since $W \ge g_*$, every sample path has at most $C_*$ renewals in any interval of length one. Thus $U$ has the same bound. If $q \ge 0$ and there is no renewal in $[q, q+H]$, some last renewal $t \le q$ is followed by a spacing longer than $q + H - t$. Independence of the next spacing gives the upper bound

$$\int_{[0,q]} \mathbb{P}(W > q + H - t)\, U(dt).$$

Partition $q - t$ into half-open unit intervals $[j, j+1)$ (assign the endpoint at zero to its containing interval). Markov's inequality bounds the integrand on the $j$th piece by $M_W e^{-\theta(H+j)}$. Summing gives $C_* M_W e^{-\theta H}/(1 - e^{-\theta})$, uniformly in $q$. For the delayed process, the event $D > v$ costs at most $M_D e^{-\theta v}$. Conditional on $D \le v$, apply the ordinary bound with $q = v - D$. This proves (10). □

For the source path take $r_{\mathrm{in}} \ge b_1 \ge 10b_0$ and

$$v = \log(r_{\mathrm{in}}/b_1), \qquad H = \log(b_1/(10b_0)).$$

A marked beginning in that time interval has pre-draw gap in $[10b_0, b_1]$. With $t \in [2.09, 2.13]$, its arrival satisfies

$$\frac{10b_0}{3.13} \le x \le \frac{b_1}{3.09}, \qquad 2.09 \le s' \le 2.13, \qquad r' > \frac{10(2.09)}{3.09} b_0 > 6b_0.$$

Thus the source's slightly larger ratio and exponent windows have failure probability at most

$$M_D(b_1/r_{\mathrm{in}})^\theta + C_\theta(10b_0/b_1)^\theta, \qquad C_\theta = \frac{C_* M_W}{1 - e^{-\theta}}. \tag{11}$$

For example, choosing $b_1/b_0 \ge 10(2C_\theta/\varepsilon)^{1/\theta}$ and $r_{\mathrm{in}}/b_1 \ge (2M_D/\varepsilon)^{1/\theta}$ makes the bound at most $\varepsilon$. The parameters $\theta, M_D, M_W$ are explicit functions of cycle transforms, not numerically certified constants for the sieve chain.

This improves the local window dependence from an unspecified finite-mean tail to an exponential tail in logarithmic width. It does not quantify the source's prime coupling error, inverse estimate, Siegel–Walfisz threshold or summed stopping comparison. All cutoff windows remain fixed before the arithmetic limit. Allowing $b_1$ to grow with $z$ would require additional uniform arithmetic estimates, not just (11).

## 7 Assembly, Boundaries, and Imported Work

The finite identities can be organized into a conditional assembly certificate without erasing the arithmetic premises. Let $Q = YV_0/B^2$. Suppose the reference root is at least $c_L Q$, the large-gap absolute defect sum is at most $\alpha Q$, the good compact defect sum is at most $\beta Q$, and all remaining retained compact defects cost at most $\gamma Q$. Suppose further that the aggregate even-stop correction is nonnegative. Then Proposition 2.2 implies

$$S_1 \ge (c_L - \alpha - \beta - \gamma)Q.$$

This is a finite implication with explicit sign and scale. The source takes three budgets at most $c_L/8$ to leave at least $5c_LQ/8$. Nonnegative aggregate correction does not say that every individual stop improves the reference value.

The global source architecture is as follows. This is a map, not a claim that every underlying proof has been independently verified.

| Source unit | Role and remaining boundary |
|---|---|
| Introduction; inputs | Arbitrary-class covering theorem; fundamental sieve lemma, prime estimates, large sieve, Kloosterman bounds and key renewal theorem. External deep inputs are imported. |
| Tree; functions | Exact smallest-bad-prime partition; delayed sieve functions; positive derivative tilt; regeneration and occupation. Finite tree algebra and the local renewal extraction are proved here. |
| Prime paths | Discrete tilted normalization, coupling with continuous paths, high-state exclusions and weighted occupation. This uniform transfer is imported, not replaced by numerical kernel compression. |
| Reference | Prime quadrature, the boundary anomaly and its signed correction, then a local estimate above ratio 2. The global positive margin is imported; its negative starting benchmark is retained. |
| Boxes | Recent witnessing edge, regular independent boxes, threshold clearance and total enlarged-box harmonic mass independent of width. Independence holds in a full box, not after conditioning on hardness. |
| Inverse | Whole-box fixed rational lists from witnessing edges, algebraic and lattice-point estimates. The growing-degree argument is not independently re-proved here. |
| Variance | Arithmetic coordinates; the independent model; small-$H$ and modular-hyperbola domination; hard endpoint height bounds. The finite random variance and its stable ledger are proved here. |
| Stops | Owners and tags, all-prime-bin averages, actual-stop comparison and capture of hard endpoints. Marked-window probability is sharpened locally; the arithmetic averaging is imported. |
| Assembly | Order of fixed parameters, positive root budget, large-prime deletion and bounded-$k$ completion. The last arithmetic sieve bound is not a consequence of finite enumeration. |
| Appendices | Modular-hyperbola completion; Buchstab limit, additive large sieve, plane intersections, and progression prime upper bound. Uninspected proofs remain explicit imported boundaries. |
| Main; route figure; references | Build structure, dependency diagram and bibliography, not additional mathematical verification. |

Three safeguards in the source deserve emphasis. First, its inverse exception fraction is measured against the whole box and its rational list is fixed before varying the isolated prime. Second, its total enlarged-box mass is $O_K(B^{-2})$, independent of sufficiently small fixed $\xi$; one must not multiply a per-box exception fraction by the number of boxes. Third, representative stop tests preserve the entire owner-aligning subset when the tag prime is replaced. That allows a full-bin average minus a bounded exceptional complement. Conditioning a prime distribution on being a favorable stop without this preservation would invalidate that step.

For the source's actual-stop comparison the reference ratio is bounded above by .33, whereas the all-bin average after removing the nonowner complement leaves .34 in the same normalization. These numbers are not numerical solutions of the whole problem: the common factor is $J_p V_{\mathrm{all}}(P)$ and the small complement uses an absolute individual upper sieve. Prime cutoffs below the last prime are strict on both sides. The source supplies the asymptotic uniformity on a fixed window before these margins are used.

The fixed-parameter order is retained: choose the compact range and error budgets, witnessing and inverse parameters, variance threshold, owner threshold and stop window, then the sufficiently small fixed bin width $\xi$, and finally $z$ large. No formula in this report licenses changing this order. In particular our $JV$ diagonal bound does not make the fundamental lemma or prime progression estimates effective in a new joint range.

# 8 Literature, Comparison, and Research Limits

A targeted primary-literature refresh was performed on October 8, 2026, using public paper titles only. The assigned source was read from the local pinned release [1]; its captured-tree identity is separate from mathematical correctness and from the state of a live branch. No exhaustive priority search is claimed.

Hajdu and Saradha's Theorem 1.2 [2] distinguishes the maximum over all prime sets from the primorial value: at 24 factors those values are 236 and 234, respectively. Their definitions and covering formulation were inspected in the primary author PDF. Ziller's primary abstract [3] reports computations of the maximum through 43 prime factors and supplies ancillary covering data. Those computations were not rerun here. Costello and Watts' primary introduction [4] explicitly uses $h(k)$ for the primorial quantity, whereas our notation follows the assigned source and takes the supremum over arbitrary prime sets. A symbol match is therefore not a theorem match.

Gallagher's work [5] supplies classical context for singular-series averaging. Its prime-count distribution conclusion uses a uniform prime-tuple conjecture; this report does not invoke that conditional conclusion. The assigned source already identifies its two-point variance argument with this classical framework. Our finite Euler cancellation is proved independently above.

Banks, Ford and Tao [6] distinguish rigorous results for randomly sieved integers from heuristic predictions about actual prime gaps. The primary arXiv metadata was checked at version 3, revised August 12, 2025; it records a correction to a lemma's error term without a change to the theorems. This report does not import that unread lemma or claim that model concentration forces a worst-case prime-class covering bound. This distinction is exactly why the source's inverse, domination and stopping arguments remain indispensable.

## 8.1 Benefits and Costs

The finite variance formula removes an asymptotic remainder from one specific source input. Its stable version isolates diagonal size, mass error, one-point error and pair error. It is useful for choosing a quantitative local exception budget. Its limitation is that the measure comparison still has to be proved arithmetically, including exactly zero pair events and frozen-interval boundaries.

Positive conditional-mean compression preserves normalization and linear moments, and gives a second-order compact error with a proved constant. Its costs are certified cell masses and first moments, tail control, and derivative bounds for the particular reward. It cannot replace all the source's discontinuous admission tests by smooth moment matching. A covariance matrix alone likewise does not determine arbitrary event probabilities.

The renewal refinement turns a qualitative choice of window ratio into a power-law choice in the original gap scale, conditional on cycle exponential transforms. It correctly permits marks to depend on cycle length. The constants still require analytic bounds for the specific continuous chain. An explicit global Jacobsthal constant would additionally require effective versions of every imported arithmetic stage and a fully quantified fixed-parameter dependency chain. That task has not been completed here.

### 8.2 Corrections and Retained Details

No substantive mathematical error in the inspected source deductions is established by this report. The finite variance estimate and exponential window bound are strengthenings of local conclusions, not repairs of false source assertions. The source already keeps the parity zero in the pair formula, uses a positive unnormalized measure without assuming prime-class independence, and preserves the dependence of a mark on its own cycle. We retain those valid details. We also retain strict versus weak prime cutoffs, the raw invariant normalization, the negative initial benchmark, and aggregate rather than pointwise stopped comparisons.

Feasible further work is narrowly defined. One may certify numerical upper bounds for $m_0(\theta), m_1(\theta)$ and the initial transform, using positive compact quadrature with explicit tails. One may also track actual $\epsilon_0, \epsilon_1, \epsilon_2, K, \delta$ through a fixed arithmetic box to obtain a numerical local bad-prime fraction. Neither program permits growing stop windows or shrinking bin widths with $z$ without new uniform estimates. A stronger global asymptotic or new prime-gap theorem is not a promised consequence.

## 9 Read Scope and Reproducible Evidence

All 15 source TeX files and the bibliography were mapped by structure. Literal bounded reading covered 4,142 of 6,741 build-text lines, including all 378 bibliography lines; the TeX-only count is 3,764 of 6,363 lines. Complete scientific reads include the introduction, inputs, tree, functions, variance and assembly. Selected reads cover the prime-path contract, reference quadrature and local comparison, compact-box and inverse contracts, and the actual-stop comparison and hard-endpoint capture. The two appendices were mapped but not literally read. Their proofs, the middle signed boundary calculation and the uninspected inverse and transfer arguments are imported, not retrospectively counted as checked.

The private scope is 358 of 1,385 lines across the two named originals. Included chapter files were not followed and no whole-corpus reading is claimed. The companion JSON ledger gives every literal interval, its unread complement and matching pre-read and post-read SHA-256 hashes. Hash agreement establishes endpoint identity, not continuous monitoring or mathematical validity.

The standalone author script `M047-exact-checks.py` uses only integer and rational arithmetic and passed 55,075 assertions. It tests 16,000 triangular identities and their nonnegative deficits; 704 complete joint-pattern models with lengths 1 through 64 over 11 prime sets; 308 nonunit progression cases; stable finite positive measures; exact node partitions and stopped comparisons; positive compact compression; and correlated marked-cycle transform and finite-window checks. The empty and odd-only prime sets are included. There are 71 tested marked-window bounds strictly below one, so that part is not solely a check of vacuous probability bounds. Counts distinguish assertions from parameter cases and are itemized in the JSON output. Parent checks are not included in the author count.

The saved finite certificate records direct counts and rational reference values for a fixed four-prime example. The scripts and their outputs are reproducible local algebra evidence, not a formal

proof assistant verification. Compilation and all-page visual inspection establish document integrity and readability, not validity of the global arithmetic theorem.

The contribution rating is C3, independent of the catalogue's L3 applicability classification. The demonstrated work consists of source-specific proved quantitative improvements: an all-length finite variance bound with measure stability and progression correlation costs, and an exponential marked-window bound in place of a qualitative finite-mean estimate. These are significant local improvements under the project's contribution scale, with classical tools and parent-supplied variance observations credited. The work is not a new solution of the quadratic Jacobsthal problem and does not certify the entire source proof.

## References


[1] OpenAI, *A quadratic bound for Jacobsthal's function*, September 25, 2026. Pinned public release. Source build and scoped readings recorded in the accompanying evidence.

[2] L. Hajdu and N. Saradha, *Disproof of a conjecture of Jacobsthal*, Mathematics of Computation 81 (2012), 2461–2471. Primary author PDF, Theorem 1.2 and covering definitions.

[3] M. Ziller, *New computational results on a conjecture of Jacobsthal*, 2019. arXiv:1903.11973v2. Abstract and version metadata inspected; ancillary computations not rerun.

[4] F. Costello and P. Watts, *An upper bound on Jacobsthal's function*, Mathematics of Computation 84 (2015), 1389–1399. Primary repository manuscript, introduction.

[5] P. X. Gallagher, *On the distribution of primes in short intervals*, Mathematika 23 (1976), 4–9. Publisher record and DOI. Classical context only; the finite identity used here is proved in full.

[6] W. Banks, K. Ford and T. Tao, *Large prime gaps and probabilistic models*, Inventiones Mathematicae 233 (2023), 1471–1518. Corrected author version 3 (2025). Primary abstract and revision metadata inspected.

[7] M. Subbarao, private original sources: `TR-core-pat-16`, lines 184–328, and `volume1-rev2/ch03-local-moments-rev2`, lines 14–226. Local confidential proof consultation; exact paths and hashes in the owned read-scope ledger.